

Ergashboyev Bobur

TashkeTECHNICAL SKILLSnt, Uzbekistan | +998 93 541 83 11 | hnot779@gmail.com

AI Software Engineer & Technical Code Evaluator Highly analytical Python Developer and AI Engineer specializing in backend architecture, LLM integrations, and algorithm optimization. Proven ability to write secure, scalable, and PEP-8 compliant Python code (FastAPI, PyTorch). Experienced in complex debugging, data structuring, and designing RAG (Retrieval-Augmented Generation) pipelines. Exceptionally detail-oriented with a strong background in assessing AI model outputs, mitigating hallucinations, and writing clear technical documentation.

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), SQL, JavaScript (Node.js).
- **AI & Machine Learning:** LLM APIs (OpenAI, Anthropic), RAG Architecture, Prompt Engineering, Output Evaluation, PyTorch, TensorFlow, CNNs.
- **Backend Engineering:** FastAPI, Django, REST API Design, Asynchronous Programming (asyncio), Webhooks.
- **Data & Infrastructure:** PostgreSQL, Supabase, SQLAlchemy, Alembic, Docker, Git, n8n Orchestration.
- **Code Quality:** Advanced Debugging, Unit Testing, Refactoring, Stack Trace Analysis, Technical Writing.

EXPERIENCE

AI / Backend Software Engineer (System Architect) *REVENANT (Enterprise AI Automation Ecosystem)*

- **LLM Evaluation & Routing:** Engineered a multi-tier AI orchestration platform integrating large language models (OpenAI, Anthropic) to handle complex, asynchronous logic processing.
- **Mitigating AI Hallucinations:** Developed a "Consensus Engine Validator" that simultaneously evaluates and compares outputs from multiple LLMs, applying strict logical conditions to ensure AI responses are mathematically and contextually accurate.
- **Backend Architecture:** Built highly resilient REST APIs using FastAPI and Python, handling dynamic rate-limiting, custom webhook integrations, and multi-stage data routing.
- **Data Integrity & Debugging:** Maintained a zero-trust PostgreSQL database architecture, utilizing complex SQL queries and strictly enforced data schemas while debugging deep asynchronous execution traces.

Backend & API Developer *Moneta (Financial Tracking API)*

- **API Development:** Architected a multi-tenant personal finance RESTful API leveraging FastAPI, SQLAlchemy, and PostgreSQL for handling concurrent transactions and relational data queries.
- **Data Normalization:** Wrote highly readable and efficient Python scripts to automatically fetch, parse, and normalize daily currency exchange data from external central bank APIs.
- **Error Monitoring & Refactoring:** Integrated Loguru and Sentry for real-time application monitoring; regularly conducted root-cause analysis on stack traces to refactor faulty logic and optimize response times.

Machine Learning Engineer (Academic Projects) *Computer Vision & Deep Learning Models*

- **Model Training:** Designed, built, and evaluated Convolutional Neural Networks (CNNs) from scratch using TensorFlow and Keras to classify image datasets.
- **Data Pipeline Construction:** Implemented comprehensive data preprocessing pipelines—cleaning, augmenting, and formatting raw data to feed into deep learning models efficiently.
- **Metric Analysis:** Extensively documented model architecture and evaluated training methodologies using strict performance metrics (loss, accuracy, validation curves).

EDUCATION

PDP University — Tashkent, Uzbekistan *Bachelor of Science in AI Solutions and Applications*

- **Current Standing:** 3rd Year (Completed 2nd Year)
- **Academic Excellence:** 4.0 / 4.0 GPA
- **Core Coursework:** Advanced Python Programming, Machine Learning, Deep Learning, Data Structures & Algorithms, Database Architecture.

Sincerely,

Ergashboyev Bobur